

Last Updated:

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General definition for **Variant A,B,C,D,E**

If $\text{PAR_ambient_temp_limit}/0 \leq \text{Amb Temp} < \text{PAR_ambient_temp_limit}/1$, then **E** = 0;

If $\text{PAR_ambient_temp_limit}/1 \leq \text{Amb Temp} < \text{PAR_ambient_temp_limit}/2$, then **E** = 1;

If $\text{PAR_ambient_temp_limit}/2 \leq \text{Amb Temp} < \text{PAR_ambient_temp_limit}/3$, then **E** = 2;

If $\text{PAR_ambient_temp_limit}/3 \leq \text{Amb Temp} < \text{PAR_ambient_temp_limit}/4$, then **E** = 3;

If $\text{PAR_ambient_temp_limit}/4 \leq \text{Amb Temp} < \text{PAR_ambient_temp_limit}/5$, then **E** = 4;

If $\text{PAR_ambient_temp_limit}/5 \leq \text{Amb Temp} < \text{PAR_ambient_temp_limit}/6$, then **E** = 5;

If $\text{PAR_ambient_temp_limit}/6 \leq \text{Amb Temp}$, then **X** = 6;

RC setting	A
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12

FC setting	B
-30	1
-28	2
-26	3
-24	4
-23	5

-22	6
-21	7
-20	8
-19	9
-18	10
-17	11
-16	12

VTC setting	C
-20	1
-18	2
-16	3
-12	4
-4	5
0	6
2	7
4	8
6	9
8	10
10	11
12	12

VFC setting	D
-20	1
-18	2
-16	3
-12	4
-4	5
0	6
2	7
4	8
6	9
8	10
10	11
12	12

Sample:

1. Temperature monitor: (hard-coded)

Read the first line of the .dat file and record the test start time, start checking the following conditions only when the test has started 6 hours. Alert if there is an anomaly.

- If TRC > 10 °C and RC doors status remains unchanged for the last 30 mins, remember to check both doors
 - TRC > 10 °C AND Door switch RC left (and right) = 0 or 1 for the past 30 minutes
- If TRC temperature is too low
 - TRC < 0 °C
- TFC > -5 °C and FC doors status remains unchanged for the last 30 mins
 - TFC > -5 °C AND Door switch FC left (and right) = 0 or 1 for the past 30 minutes

2. Fan behavior monitor: (hard-coded)

If any of the following conditions is met, alert should be activated:

- Compressor RPM is >0, AND FC valve =1 , AND FC fan duty = 0 for more than 3 minutes.
- Compressor RPM is >0, AND VFC valve =1 , AND VFC fan duty = 0 for more than 3 minutes.
- FC fan duty = 0 last for more than 5hs
- VFC fan duty = 0 last for more than 5hs

Normal mode:

- FC valve = 1, AND FC fan duty not equal to PAR_FC_fan_initial_speed_supply/E
- VFC valve = 1, AND VFC fan duty not equal to PAR_VF_fan_initial_speed_supply/E
- Compressor RPM is >0, AND Amb Temp > 20C, AND Cond fan duty = 0 for more than 3 minutes
- FC valve = 1, AND Cond fan duty not equal to PAR_FC_condenser_fan_initial_speed_supply /E
- VFC valve = 1, AND Cond fan duty not equal to PAR_VF_condenser_fan_initial_speed_supply /E

3. Compartment flap behavior monitor:

If any of the following conditions is met, alert should be activated:

- flap open = 0 or 1 for the past 5hs

- If Temp VTC > **C** + PAR_CC_ambient_related_setpoint_temp_offset_ **C/E**
And flap open = 0

4. Stop valve behavior monitor:

If any of the following conditions is met, alert should be activated:

- **D** If Temp VFC > **D** + PAR_RC_ambient_related_setpoint_temp_offset_ **C/E**
And VFC valve = 0

5. Compressor behavior monitor:

If any of the following conditions is met, alert should be activated:

- VFC valve = 1 And Compressor RPM = 0
- FC valve = 1 And Compressor RPM = 0